Development of a Machine Learning model to classify different postures for the sitting position

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Abstract

This paper presents the development of a Machine Learning model aimed at classifying sitting postures, with the broader goal of promoting ergonomic practices and preventing musculoskeletal disorders in the workplace. The motivation behind this study lies in the growing concern over health issues related to prolonged incorrect postures during desk work. Data were collected using pressure sensors embedded in a prototype chair for testing purposes, capturing ten distinct sitting postures, each sustained for 30 seconds. The data analysis was conducted on the Google Colab platform with the PyCaret library, enabling efficient testing and comparison of five supervised classification algorithms: Random Forest (RF), eXtreme Gradient Boosting (XGB), Artificial Neural Networks (ANN), Support Vector Machines (SVM), and k-Nearest Neighbors (KNN). The best performance was achieved with the XGB model, particularly when trained with filtered pressure images and Body Mass Index (BMI) as an additional variable, reaching an F1-Score of 0.9750. In parallel, correlation analysis was conducted between variables such as average pressure, maximum pressure, area of contact, and percentage of pressure, providing insights into variable redundancy and complementarity across postures. The findings highlight the feasibility of integrating intelligent posture-monitoring systems into everyday office environments, contributing to the advancement of workplace ergonomics and proactive health monitoring.

Keywords: Machine Learning, Pressure Maps, Ergonomics, Posture Classification, Pressure Sensors, Occupational Health

1 Introduction

Adopting inappropriate postures for long periods of sitting is a significant factor in the development of musculoskeletal disorders (MSDs), that may compromise the quality of life and performance in the workplace. These disorders represent a growing concern in various areas, especially those involving prolonged sedentary activities, such as offices and operation centers [1].

In recent years, advances in technologies based on pressure sensors, combined with machine learning algorithms, have enabled new approaches to postural monitoring [2]. These solutions offer the potential to identify patterns of postural behavior in real time, enabling preventive and corrective interventions.

In this context, this work proposes the development of a computational model for the automatic classification of different sitting postures, based on data obtained by pressure sensors. These sensors are integrated in a prototype instrumented chair, developed by an industrial partner.

The main objective is to analyze the suitability of using different machine learning algorithms to identify user posture in the prototype system. Therefore, aspects such as identification accuracy and suitability for real-time operation will be considered.

By fulfilling this objective, the prototype may evolve into a system that can contribute to the prevention of musculoskeletal injuries and diseases, optimization of ergonomic practices and, consequently, improvement of physical well-being in the workplace [3].

The methodology adopted in this study involves collecting postural data using pressure sensors installed in a chair, followed by the application of machine learning techniques to create and evaluate classification models.

This article is organized as follows: Section 2 presents a review of the state of the art on postural monitoring and machine learning applications in this context. Section 3 describes the methodology used, including data collection, model development, and evaluation procedures, providing a comprehensive overview of the experimental workflow. Section 4 presents the results obtained. Section 5 discusses these results in detail. Finally, Section 6 outlines the conclusions and suggestions for future work.

2 State of the art

The increasing prevalence of sedentary lifestyles, particularly in occupational settings, has heightened concerns regarding the health implications of prolonged sitting. Poor sitting posture is a significant contributor to musculoskeletal disorders (MSDs), which can lead to chronic pain and decreased quality of life. To address these issues, researchers have explored various technologies for monitoring and improving sitting posture.

2.1 Sensor-Based Posture Monitoring

Among the technologies investigated, pressure sensor-based systems have gained prominence due to their non-intrusive nature and ability to capture detailed pressure distribution data. These systems typically involve embedding pressure sensors within seating surfaces to monitor the user's posture in real-time. For instance, the Sitting Posture Recognition System (SPRS) utilizes a matrix of pressure sensors to classify sitting postures with an accuracy of up to 99.1%, demonstrating the potential of such systems in real-world applications [4] [5].

Additionally, smart sensing chairs equipped with Force Sensing Resistors (FSRs) have been developed to detect and classify various sitting postures. These chairs leverage machine learning algorithms, such as Convolutional Neural Networks (CNNs) and Artificial Neural Networks (ANNs), to analyze pressure data and provide feedback to users. However, challenges remain in achieving high classification accuracy, particularly due to limitations in training data diversity [6].

2.2 Machine Learning Approaches

Machine learning techniques have been extensively applied to enhance the accuracy and efficiency of posture classification systems. A comprehensive literature review highlights the utilization of various algorithms, including Support Vector Machines (SVMs), k-Nearest Neighbors (k-NN), Decision Trees, and Random Forests, in classifying sitting postures based on pressure sensor data

[4]. These algorithms have shown promising results, with some systems achieving classification accuracies exceeding 90% [7].

Furthermore, advancements in deep learning have introduced more sophisticated models capable of handling complex and high-dimensional data. For example, the integration of Spiking Neural Networks (SNNs) with pressure sensor data has demonstrated improved performance in recognizing subtle posture variations [5]. Such approaches offer the potential for more adaptive and personalized posture monitoring solutions [6].

2.3 Challenges in Postural Identification

Despite the progress in sensor technologies and machine learning algorithms, several challenges persist in the development of effective posture monitoring systems [6]. Key issues include sensor placement optimization, data variability among users, efficient real-time operation, and the need for large, diverse datasets to train robust models. Moreover, ensuring user comfort and system unobtrusiveness remains critical for widespread adoption [8].

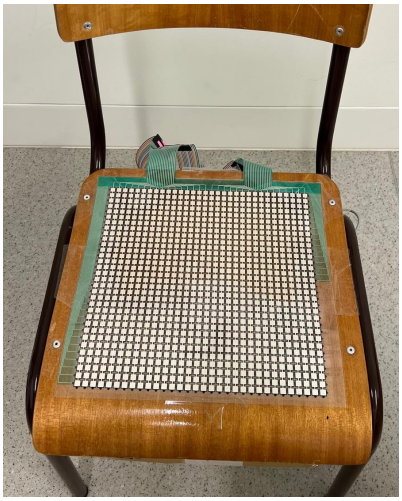
3 Methodology

This section outlines the experimental procedures, data acquisition strategy, and modeling approach used in this study. A systematic framework was adopted to ensure consistent data collection and to evaluate the performance of various machine learning algorithms in classifying seated postures based on pressure sensor input.

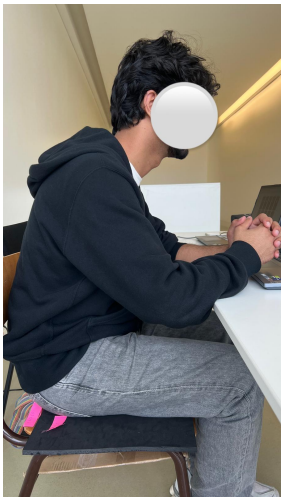
3.1 Data Collection

Following approval from the Ethics Committee of the Polytechnic Institute of Coimbra, a total of 40 volunteers participated in the study (Figure 1b), helping collect data using a prototype instrumented chair equipped with pressure sensors strategically distributed across the seating surface (Figure 1a). These sensors were connected to a pressure mapping system, integrated with an arduino and supported by a customizable software, which enabled both real-time visualization and continuous recording of the data.

Each volunteer was instructed to simulate 10 distinct sitting postures. Each posture was held for a duration of 30 seconds. The postures selected focus on pressure-based ergonomic assessment for seated configurations (Figure 2).



(a) Instrumented chair setup.



(b) Example of participant during data collection.

Figure 1: Experimental setup: (a) pressure-sensor chair; (b) data collection with participant.



Figure 2: Set of 10 simulated sitting postures adopted by participants [9].

3.2 Model Development and Evaluation

The classification pipeline was developed using the PyCaret machine learning framework within a Google Colab environment, offering a cloud-based infrastructure that ensured both computa-

tional efficiency and experimental reproducibility. This framework streamlined the processes of data preprocessing, feature engineering, model training, and performance evaluation, allowing for rapid prototyping and consistent comparisons across multiple algorithms. The raw data collected from the pressure sensors consisted of time-series matrices representing force distribution (in kg/cm^2) across a 32×32 grid, along with additional variables such as average pressure, maximum pressure, center of pressure (CoP), percentage of total pressure per quadrant, and contact area.

As part of the preprocessing stage, a filtering procedure was applied to reduce high-frequency noise caused by small involuntary body movements and sensor fluctuations. This filter consisted of a median filter applied frame by frame across the pressure maps, replacing each cell's value by the median of its 3×3 neighborhood. This operation preserved edges while smoothing out isolated spikes, resulting in more stable input patterns for the classification algorithms.

A total of five supervised classification algorithms were evaluated in this study: Random Forest (RF), eXtreme Gradient Boosting (XGB), Support Vector Machine (SVM), Artificial Neural Networks (ANN), and k-Nearest Neighbors (KNN). These algorithms were selected due to their demonstrated effectiveness in prior works involving sensor-based classification tasks.

Each model was trained under multiple configurations, using both raw and preprocessed pressure data. Additionally, the inclusion of various features tested. These are the Body Mass Index (BMI), Center of Pressure (CoP), and Percentage of Pressure distributed across the sensing matrix.

To ensure robust generalization and minimize the risk of overfitting, all models were evaluated using stratified 10-fold cross-validation. This approach allowed each posture class to be proportionally represented in both training and validation folds, enhancing the reliability of the evaluation process.

Performance was assessed using two primary metrics: accuracy and F1-score. Accuracy provides a global view of classification correctness across all posture classes, while F1-score accounts for class imbalance and offers a harmonic mean of precision and recall. This combination was chosen to ensure a balanced evaluation, particularly important in multiclass problems such as seated posture classification, where certain postures may be harder to distinguish due to subtle variations in pressure distribution.

4 Results

able 1 presents the accuracy results of the five evaluated models across eight dataset configurations. These configurations include combinations of raw sensor data, median-filtered data, and engineered features such as Body Mass Index (BMI), Center of Pressure (CoP), and Percentage of Pressure (%Press). This consolidated table enables a direct comparison of model behavior across all experimental conditions.

Among all models, ANN consistently achieved high performance, with its best result (0.9486) observed in the "No Filter + BMI" configuration. This suggests that the model is particularly effective when using raw pressure data augmented with simple physiological features. Even in the absence of any engineered variables, ANN maintained an excellent baseline accuracy of 0.9463 in the "No Filter" setting. While filtering slightly reduced its performance (e.g., 0.9379 in "Filter"), the model remained robust across all conditions, never falling below 0.9355.

Random Forest (RF) also showed strong and stable performance across all configurations. It achieved its highest accuracy of 0.9475 in the "No Filter + BMI" setup, closely followed by 0.9415 in "No Filter + %Press". These results suggest that RF benefits from engineered features and performs well both with and without filtering.

XGBoost (XGB) clearly outperformed all other models in terms of raw accuracy, reaching a peak value of 0.9761 in the "Filter + BMI" configuration. It also performed very well with "No Filter + BMI" (0.9677), demonstrating its ability to capitalize on informative engineered features. However, its performance declined slightly when using CoP or %Press, indicating that BMI is a more discriminative feature in this context.

In contrast, Support Vector Machine (SVM) showed high accuracy with raw unfiltered data (0.9390), but its performance dropped drastically when engineered features were introduced. It fell to 0.2186 with BMI and as low as 0.0454 with %Press. This suggests that SVM is highly sensitive to feature scaling and transformations, and may struggle to generalize in more complex or heterogeneous feature spaces.

K-Nearest Neighbors (KNN) consistently exhibited the lowest classification accuracy among all models. It performed best in the "No Filter + BMI" condition (0.7193), but its accuracy dropped significantly in other setups, particularly when CoP or %Press features were introduced. The poor performance of KNN indicates that it is not well suited for this high-dimensional and pressure-based dataset without substantial optimization.

In summary, ANN and RF proved to be the most balanced and robust models across configurations, while XGB showed superior accuracy when properly tuned with filtered data and BMI. Meanwhile, SVM and KNN revealed substantial limitations, especially in handling engineered features and complex data transformations.

Table 1: Models Accuracy across various dataset configurations

Configurations	**RF**	**XGB**	**ANN**	**KNN**
Filter + BMI	0.9391	0.9701	0.9427	0.6727
No Filter + COP	0.9379	0.9164	0.9451	0.4387

4.1 Model Performance Based on F1-Score

Table 2 presents the F1-Scores obtained by the evaluated models across eight different dataset configurations, which combine raw sensor data, median filtering, and engineered features including Body Mass Index (BMI), Center of Pressure (CoP), and Percentage of Pressure (%Press). This comprehensive view highlights the models' ability to handle both raw and enriched data sources in posture classification tasks.

Artificial Neural Networks (ANN) emerged as the most consistently high-performing model. Its best F1-score, 0.9449, was achieved in the "No Filter" configuration using only raw data. Even when engineered features or filtering were introduced, ANN maintained excellent performance, never falling below 0.9305. For instance, it achieved 0.9434 with "No Filter + BMI" and 0.9404 with "No Filter + CoP", confirming its robustness and adaptability across different feature representations. The stability and high recall demonstrated by ANN make it well-suited for real-world implementations in dynamic environments.

Random Forest (RF) also demonstrated strong and reliable performance. It reached a peak F1-score of 0.9430 in the "No Filter + BMI" configuration and maintained values above 0.925 in every setting, including more complex feature combinations. Notably, RF scored 0.9361 with "No Filter + %Press" and 0.9326 with "No Filter + CoP", confirming its resilience to changes in data preprocessing and feature composition. These results reinforce RF's practicality for scenarios demanding efficient, interpretable models with stable behavior across varying input conditions.

XGBoost (XGB) delivered the highest overall F1-score, achieving 0.9750 in the "Filter +

BMI" configuration. This underscores its strength in leveraging structured and stable features, especially when median filtering is applied to reduce noise. XGB also performed very well in "No Filter + BMI" (0.9651), suggesting it is particularly effective when provided with anthropometric features like BMI. Although its performance dipped slightly in %Press and CoP-based configurations (ranging from 0.9006 to 0.9096), it consistently outperformed most models in these setups as well. Thus, XGB excels in conditions where preprocessing and feature engineering are feasible.

In contrast, Support Vector Machine (SVM) showed high variance in its performance. While it attained a competitive 0.9343 with raw data, the inclusion of engineered features led to substantial degradation. F1-scores fell dramatically to 0.1580 with "No Filter + BMI", 0.1149 with "Filter + BMI", and even lower with pressure-derived features—dropping to 0.0205 with "Filter + %Press". Although one configuration with filtered CoP yielded a respectable 0.8842, the overall instability suggests SVM's sensitivity to non-linearly distributed or colinear features, limiting its reliability in heterogeneous data environments.

K-Nearest Neighbors (KNN) consistently underperformed compared to the other models. Its highest F1-score, 0.6999, was recorded with "No Filter + BMI", showing some benefit from the addition of anthropometric data. However, all other configurations resulted in scores below 0.5—falling as low as 0.3053 with "Filter + %Press". These outcomes reflect KNN's inherent limitations in managing high-dimensional, noisy data, and its poor scalability in multiclass classification with engineered features.

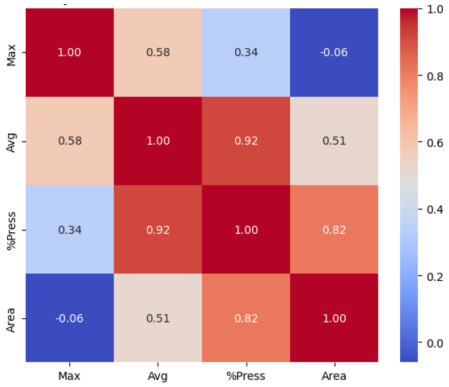
In summary, ANN and RF are the most robust and versatile models across all tested conditions, offering high performance without over-dependence on specific preprocessing techniques. XGBoost stands out for its top-tier accuracy under optimal configurations but requires careful feature engineering and filtering. Meanwhile, SVM and KNN showed severe limitations, reinforcing the need for tailored model selection depending on feature complexity and deployment constraints.

Table 2: Models F1-Score across various dataset configurations

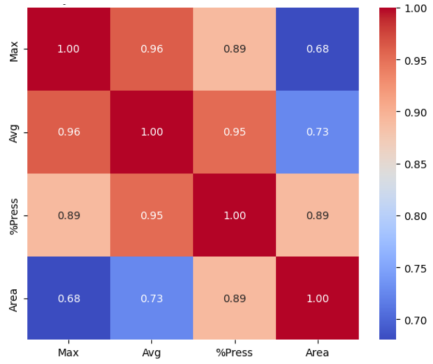
Configurations	**RF**	**XGB**	**ANN**	**KNN**
Filter + BMI	0.9332	0.9750	0.9371	0.6551
No Filter + COP	0.9326	0.9089	0.9404	0.4207

4.2 Correlation Analysis Between Pressure-Derived Features

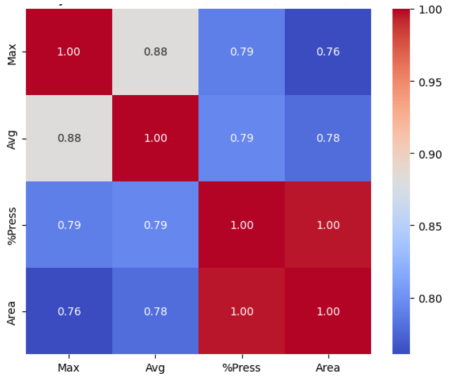
To better understand the relationships between the pressure-derived varieds used in the classification, a correlation analysis was performed for three postures: P3 (forward-leaning), P7 (back supported, right leg crossed), and P10 (sitting on the edge of the seat). The corresponding correlation matrices are presented in Figures 3a, 3b, and 3c.



(a) Posture P3 – Leaning forward



(b) Posture P7 – Back supported, right leg crossed



(c) Posture P10 – Sitting on the front edge

Figure 3: Correlation matrices between pressure-derived features for selected postures: P3, P7, and P10.

For posture P3 (Figure 3a), the correlation between maximum pressure and contact area was –0.06, which is likely not statistically significant. This suggests that there is no meaningful correlation between these two features, meaning they vary independently across participants. Such lack of association may reflect a more localized pressure distribution—typically in the front of the seat—leading to inter-subject variability that can hinder classification accuracy, especially in models that rely on global spatial patterns.

The features examined in these correlation matrices were:

- Maximum pressure: the highest value recorded across all sensors at a given time;
- Average pressure: the mean pressure value over all active sensors;
- %Pressure: the percentage of total pressure applied within a specific region (quadrant-based);
- Area: the total number of sensors registering pressure above a set threshold, representing the contact surface.

These four features were chosen as representative pressure-derived variables, commonly used in the literature for characterizing postural load distribution. They are computationally efficient, interpretable, and relevant across a wide range of ergonomic monitoring scenarios.

In contrast, posture P7 (Figure 3b) exhibited strong internal correlations among all variables. Maximum, average, and %Pressure values were highly interrelated, reflecting a uniform and stable pressure pattern. This likely results from the natural support of the chair back and leg-crossing, which reduce variability and facilitate accurate classification.

Posture P10 (Figure 3c)—where subjects sat on the front edge of the seat—showed moderate to high correlations, especially between area and %Pressure. Despite the stronger overall consistency, this posture remained sensitive to minor shifts in weight and body placement, which may explain reduced performance in models such as SVM and KNN.

In summary, these feature correlation analyses underscore how internal consistency among pressure-derived metrics can impact model performance. Postures with coherent pressure signatures (like P7) tend to produce higher classification accuracy, while postures with fragmented or unstable signatures (like P3) introduce more classification noise. This highlights the importance of context-aware feature selection in future model iterations.

5 Discussion

From an implementation standpoint, the results reinforce the value of relative pressure metrics, particularly BMI, which consistently improved classification performance while being computationally simple to compute. The inclusion of engineered features proved beneficial for certain models—especially XGBoost and Random Forest—while detrimental for others, such as SVM, which showed notable sensitivity to transformed or composite variables.

Filtering was also assessed as part of the preprocessing strategy. A median filter was applied to the raw sensor matrices to reduce noise caused by involuntary micro-movements or brief fluctuations in contact pressure. While this smoothing technique led to modest gains in classification consistency for some postures—particularly those with dynamic weight shifts like P10—it did not consistently yield statistically significant improvements in overall accuracy across models. For example, ANN showed slightly reduced performance when filtering was applied, suggesting that in some cases, important discriminative details might have been smoothed out.

Furthermore, the feature configurations tested in this study did not include all three engineered features (BMI, Center of Pressure (CoP), and Percentage of Pressure) simultaneously. This decision was made to avoid multicollinearity and retain experimental clarity. Initial correlation analyses showed that BMI and %Press, as well as %Press and Area, exhibited strong interdependence. Including all engineered variables together might have introduced redundant information, compromising both model generalization and interpretability. Additionally, testing combinations in isolation allowed for a more granular understanding of each feature's individual impact on classification performance.

The correlation analysis further revealed that postures with high internal feature coherence, such as P7, were consistently classified with high accuracy. In contrast, postures with more localized or variable pressure signatures, such as P3 and P10, exhibited lower consistency across models. This highlights the need for posture-specific strategies, potentially including tailored models, context-aware features, or adaptive thresholds for filtering and feature selection.

In practical workplace scenarios where continuous posture monitoring is needed, achieving a balance between technical accuracy, computational efficiency, and deployability is crucial. This study demonstrates that effective seated posture classification is feasible using low-cost sensor arrays and open-source ML frameworks, but successful real-world implementation requires

thoughtful system architecture and resource-aware deployment strategies.

In that context, Random Forest emerged as a leading candidate for real-time integration, given its fast inference speed, low computational overhead, and tolerance to noise. Artificial Neural Networks (ANN) also performed robustly, particularly in embedded scenarios where minimal preprocessing is preferred. These models offer promising throughput and maintain both robustness and interpretability.

On the other hand, XGBoost, while delivering the highest accuracy in filtered configurations with engineered features, demands significant computational resources and hyperparameter tuning, making it more suitable for offline batch analysis or hardware-accelerated systems. KNN, despite its conceptual simplicity, was hindered by memory demands and slow prediction times, limiting its scalability. SVM proved highly sensitive to feature composition and scale, which reduced its reliability in dynamic, heterogeneous environments.

Overall, these findings validate the feasibility of implementing posture classification systems that are not only technically robust but also operationally viable, paving the way for real-time ergonomic monitoring with concrete implications for occupational health and safety.

6 Conclusion and Future Work

This study proposed and evaluated a machine learning-based system for the classification of seated postures using pressure sensor data and derived features. A comprehensive set of experiments was conducted using five supervised learning algorithms—Random Forest, XGBoost, Support Vector Machine, Artificial Neural Networks, and k-Nearest Neighbors—applied to multiple feature configurations. The results demonstrated that models such as ANN and RF consistently achieved high classification accuracy and F1-score, particularly when using raw data or percentage-based pressure features.

The correlation analysis further revealed that postural stability and feature interdependence strongly influence classification performance. Postures with high internal feature correlation (e.g., P7) were more easily identified, while those with varied pressure concentration (e.g., P3 and P10) posed greater challenges. These insights reinforce the importance of thoughtful feature engineering and posture-specific analysis in sensor-based ergonomic systems.

From a practical standpoint, the discussion in Section 5 highlights that by selecting the appropriate machine learning algorithm, accurate posture recognition is achievable using relatively simple hardware and open-source tools. Moreover, given the lightweight nature of the trained models — particularly RF and ANN — the system shows strong potential for real-time integration. In future work, these models may be implemented in C# environments, enabling seamless integration with existing ergonomic monitoring systems. Such implementations could support live classification, while also computing performance metrics such as accuracy and F1-score in real time, allowing for adaptive feedback and system self-assessment during data acquisition sessions.

Future research should explore model generalization across diverse populations and extended sitting durations, as well as the inclusion of temporal features and sensor fusion strategies. Additionally, the deployment of this system in workplace environments will allow further validation of its usability, responsiveness, and ergonomic impact under real-world conditions.

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